US–Iran Conflict Analysis Dashboard (2026)

Project Documentation

Digital Egypt Pioneers Initiative – Graduation Project

Project Name	ConflictScope – US–Iran Conflict Data Analysis Dashboard
Track	Data Analysis
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GitHub Repository	https://github.com/darealbomo7/US-Iran-War-Data-Analysis

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1. Project Planning & Management

1.1 Project Overview

ConflictScope is a data analysis project designed to analyze and visualize data related to the 2026 US–Iran conflict.

The project aims to transform raw conflict-related data into meaningful insights through data cleaning, exploratory data analysis (EDA), statistical analysis, and interactive visualization.

The dashboard will provide an overview of military events, reported casualties, geographical distribution of conflict incidents, humanitarian impacts, and potential economic effects.

The project will use publicly available datasets and reports from reliable sources, including IranWar.ai, ACLED, and the United Nations.

Python will be used for data collection, cleaning, preprocessing, and analysis, while Power BI will be used to create an interactive dashboard that presents the findings in a clear and accessible way.

1.2 Project Objectives

The main objectives of ConflictScope are:

1. Collect and integrate publicly available data related to the US–Iran conflict.
2. Clean and preprocess raw datasets to ensure consistency and reliability.
3. Analyze the frequency and distribution of military events over time.
4. Examine reported fatalities and injuries across affected countries and regions.
5. Identify geographical areas most affected by conflict-related incidents.
6. Analyze reported damage to civilian infrastructure and humanitarian impacts.
7. Explore potential relationships between conflict developments and economic indicators, such as oil prices.
8. Develop an interactive Power BI dashboard to communicate analytical findings.
9. Provide data-driven insights that help users better understand the conflict and its consequences.

1.3 Project Scope

In Scope

The project will focus on:

- Military events and reported attacks.
- Reported casualties and injuries.
- Geographical distribution of conflict incidents.
- Types and frequency of conflict events.
- Reported damage to civilian infrastructure.
- Humanitarian consequences.
- Economic indicators, including oil price movements.

- Interactive data visualization and dashboard development.

Out of Scope

The project will not:

- Predict future military operations or attacks.
- Provide classified or confidential military information.
- Verify military claims independently through field investigations.
- Establish causal relationships between the conflict and economic changes.
- Provide political opinions or military recommendations.

The analysis will be limited to publicly available information and the coverage of the selected datasets.

1.4 Project Methodology

The project will follow these main phases:

Phase 1: Data Collection

- Identify reliable data sources.
- Download relevant datasets and reports.
- Collect conflict events, casualty records, and economic indicators.

Phase 2: Data Cleaning and Preparation

- Handle missing and inconsistent values.
- Remove duplicate records where appropriate.
- Standardize dates, country names, and event categories.
- Check data types and formats.
- Document data quality limitations.

Phase 3: Exploratory Data Analysis (EDA)

- Calculate descriptive statistics.
- Analyze event trends over time.
- Examine casualty distributions.
- Compare geographical patterns.
- Identify important trends and unusual observations.

Phase 4: Data Visualization

- Create charts using Python.
- Develop geographical visualizations.
- Design interactive Power BI dashboards.

Phase 5: Documentation and Presentation

- Document the analysis process.
- Summarize findings and limitations.
- Prepare the final project presentation.
- Upload project files and documentation to GitHub.

1.5 Project Timeline

Phase	Activities	Duration
Phase 1	Project planning and research	Week 1
Phase 2	Data collection	Week 2
Phase 3	Data cleaning and preprocessing	Week 3
Phase 4	Exploratory data analysis	Week 4
Phase 5	Data visualization	Week 5
Phase 6	Power BI dashboard development	Week 6
Phase 7	Testing and documentation	Week 7
Phase 8	Final presentation and submission	Week 8

1.6 Team Roles and Responsibilities

The team will distribute responsibilities according to members' skills and project requirements.

Role	Responsibilities
Project Manager	Coordinate activities, monitor progress, and ensure deadlines are met.
Data Researcher	Identify and collect reliable datasets and supporting reports.
Data Analyst	Perform data cleaning, preprocessing, and exploratory analysis using Python.
Data Visualization Specialist	Develop charts and interactive dashboards using Power BI.
Documentation Specialist	Prepare project documentation and organize project files.

Team members may collaborate across multiple responsibilities depending on the team size.

1.7 Risk Management

Potential Risk	Impact	Mitigation Strategy
Missing data	Incomplete analysis	Use multiple reliable data sources where possible.
Inaccurate or conflicting casualty reports	Misleading results	Cross-check figures and document their sources and uncertainty.
Duplicate records	Incorrect calculations	Apply appropriate data validation and deduplication techniques.

Potential Risk	Impact	Mitigation Strategy
Limited data availability	Reduced analytical scope	Use alternative public datasets and reports.
Time management issues	Project delays	Establish milestones and monitor progress regularly.
Technical difficulties	Delayed development	Use documented libraries and maintain backup project files.
Different data formats	Integration difficulties	Standardize formats during preprocessing.

1.8 Key Performance Indicators (KPIs)

The success of the project will be evaluated using the following indicators:

- Number of relevant datasets collected and integrated.
- Percentage of records successfully processed.
- Accuracy and consistency of data transformations.
- Completion of the planned analytical tasks.
- Number of meaningful visualizations developed.
- Successful implementation of interactive dashboard filters.
- Completion of project documentation.
- Successful testing and delivery of the final dashboard.

2. Literature Review

2.1 Introduction

This section reviews existing platforms, datasets, and analytical solutions related to armed conflicts and humanitarian crises.

The purpose is to understand how existing systems collect, organize, analyze, and present conflict-related information and identify opportunities for developing ConflictScope.

2.2 Existing Solutions and Data Sources

2.2.1 ACLED (Armed Conflict Location & Event Data)

ACLED is an organization that collects and analyzes data about political violence, demonstrations, and conflict events worldwide.

Main Features:

- Records conflict events with dates and geographical locations.
- Provides event classifications.
- Includes reported fatalities.
- Offers geographical and temporal analysis.
- Provides downloadable datasets.

Advantages:

- Structured event-level data.
- Extensive geographical coverage.
- Regular data updates.
- Useful for statistical and geographical analysis.

Limitations:

- Detailed data access may require registration.
- Reported fatalities may be uncertain or disputed.
- Data coverage depends on available reporting sources.

Relevance to ConflictScope:

ACLED can provide structured conflict-event records that support geographical analysis, event classification, and casualty comparisons.

Website: https://acleddata.com/

2.2.2 IranWar.ai Research Dataset

IranWar.ai provides a research-oriented dataset focused on the 2026 US–Iran conflict.

Main Features:

- Conflict-related event records.
- Military developments.

- Reported casualties.
- Humanitarian information.
- Economic and diplomatic indicators.

Advantages:

- Focuses specifically on the US–Iran conflict.
- Combines different dimensions of the conflict.
- Provides structured data suitable for analytical projects.

Limitations:

- Its available historical release covers a limited period.
- Some information may be based on preliminary or AI-assisted reporting.
- Data requires validation and comparison with independent sources.

Relevance to ConflictScope:

This dataset will serve as one of the main data sources for the project, providing a foundation for event analysis and visualization.

Website: https://github.com/jethomasphd/WarTheater

2.2.3 United Nations (UN) and OCHA Reports

The United Nations and its humanitarian coordination office publish reports about humanitarian emergencies and their consequences.

Main Features:

- Reports on civilian casualties.
- Information about displaced populations.
- Infrastructure damage assessments.
- Humanitarian needs and emergency responses.

Advantages:

- Provides humanitarian context.
- Includes information from international organizations.
- Helps explain the human impact of conflicts.

Limitations:

- Information is often published in PDF reports rather than structured datasets.
- Reports may have different publication dates and coverage periods.
- Some figures are estimates and may be revised.

Relevance to ConflictScope:

UN reports will be used as supplementary sources to provide context and validate humanitarian indicators where comparable data is available.

Website: https://www.un.org/

2.2.4 GDELT Project

GDELT is a global database that monitors news media and records information about events, people, organizations, and locations.

Main Features:

- Monitors international news coverage.
- Provides event-related information.
- Supports geographical and temporal analysis.
- Offers access to large-scale datasets.

Advantages:

- Broad international coverage.
- Useful for identifying media-reported trends.
- Provides machine-readable data.

Limitations:

- News coverage does not represent all real-world events.
- Multiple reports may describe the same incident.
- Automated classifications may contain errors.

Relevance to ConflictScope:

GDELT can potentially provide supplementary information for studying media-reported conflict trends and comparing reporting patterns.

Website: https://www.gdeltproject.org/

2.3 Comparative Analysis

Feature	ACLED	IranWar.ai	UN/OCHA	GDELT
Conflict Events	Yes	Yes	Limited	Yes
Casualty Information	Yes	Yes	Yes	Limited
Geographical Information	Yes	Yes	Yes	Yes
Humanitarian Information	Limited	Yes	Yes	Limited
Economic Information	Limited	Yes	Limited	Possible
Structured Data	Yes	Yes	Limited	Yes

2.4 Research Gap and Project Contribution

Existing platforms provide valuable information about conflict events, casualties, and humanitarian impacts. However, the information is distributed across multiple platforms and presented in different formats.

Some sources focus primarily on recording conflict events, while others emphasize humanitarian reporting or news coverage.

ConflictScope aims to bring selected publicly available information together into a unified analytical dashboard.

The project's main contribution is to provide:

1. A consolidated analytical view of selected conflict-related indicators.
2. Interactive visualizations of military events and reported casualties.
3. Geographical and temporal analysis.
4. A supplementary view of humanitarian and economic indicators.
5. Clear documentation of data sources, definitions, and limitations.

The project will not replace existing research platforms. Instead, it will provide an accessible analytical layer that combines selected information from these sources.

2.5 References

1. ACLED. Armed Conflict Location & Event Data. https://acleddata.com/
2. IranWar.ai Research Dataset. WarTheater GitHub Repository. https://github.com/jethomasphd/WarTheater
3. United Nations. https://www.un.org/
4. United Nations Office for the Coordination of Humanitarian Affairs. https://www.unocha.org/
5. GDELT Project. https://www.gdeltproject.org/

3. Requirements Gathering

3.1 Introduction

Requirements gathering identifies the needs of the project's stakeholders and defines the features and quality standards that ConflictScope must provide.

The objective is to ensure that the final solution is useful, understandable, reliable, and easy to interact with.

3.2 Stakeholders

The main stakeholders include:

1. General Users and Researchers

Individuals interested in understanding conflict developments through data and visualizations.

Needs:

- Access to organized conflict information.
- Clear visual representations.
- Easy comparison between different countries and periods.
- Reliable references for reported information.

2. Data Analysts

Individuals interested in exploring conflict-related datasets and identifying trends.

Needs:

- Clean and structured datasets.
- Accurate calculations.
- Filtering and comparison capabilities.
- Exportable analytical results where supported.

3. Humanitarian Organizations and Researchers

Organizations and researchers interested in understanding the humanitarian consequences of conflict.

Needs:

- Information about reported casualties and injuries.
- Geographical distribution of affected areas.
- Available information about infrastructure damage.
- Clear explanations of data limitations.

4. Academic Institutions and Students

Students and researchers who need to understand conflict data through analytical methods.

Needs:

- Accessible visualizations.

- Documented data sources.
- Clear analytical methodology.
- Reproducible results where possible.

3.3 Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement	Description
FR-01	Data Collection	The system must use publicly available conflict-related datasets.
FR-02	Data Cleaning	The system must handle missing values, duplicates, and inconsistent formats.
FR-03	Data Integration	The system should combine compatible information from different sources.
FR-04	KPI Calculation	The system must calculate key indicators such as recorded events and reported casualties.
FR-05	Event Analysis	The system must display conflict events over time.
FR-06	Casualty Analysis	The system must visualize reported fatalities and injuries where available.
FR-07	Geographical Analysis	The system should display the geographical distribution of recorded events.
FR-08	Event Classification	The system must categorize events according to available event types.
FR-09	Humanitarian Analysis	The system should present available information about civilian impacts and infrastructure damage.
FR-10	Economic Analysis	The system should visualize selected economic indicators, such as oil price movements.
FR-11	Interactive Filtering	Users must be able to filter dashboard results by available dimensions, such as date and country.
FR-12	Data Visualization	The system must present results through charts, tables, and geographical maps.

3.4 Non-Functional Requirements

Non-functional requirements describe how the system should perform.

1. Performance

- Dashboard pages should load within a reasonable time.
- Data processing should be efficient.
- Visualizations should respond smoothly to user interactions.

2. Accuracy and Reliability

- Calculations must be based on validated data.
- Data sources must be documented.
- Conflicting casualty figures must be clearly identified.

- Missing information must not be treated as zero.
- Results must distinguish between confirmed figures and estimates.

3. Usability

- The dashboard should have a clear and simple interface.
- Charts should include meaningful titles and labels.
- Filters should be easy to understand.
- Information should be presented in an organized manner.

4. Security

- The project should not expose private or confidential information.
- Only publicly available data should be used.
- Project files and source code should be maintained securely.

5. Maintainability

- Python scripts should be organized and documented.
- Data processing steps should be reproducible.
- The project structure should allow future updates and improvements.

6. Compatibility

- The dashboard should be accessible through supported Power BI viewing environments.
- Data files should use standard formats such as CSV and Excel.

3.5 Dashboard Requirements and Main Outputs

The final dashboard will contain the following analytical components:

1. Key Performance Indicators (KPIs)

The dashboard will display:

- Total Recorded Conflict Events.
- Total Reported Fatalities.
- Total Reported Injuries.
- Number of Affected Countries.

These values will be calculated from the selected dataset and its available coverage.

2. Conflict Timeline

A time-series visualization showing the development of recorded conflict events over time.

Purpose: Identify periods with increased or decreased event frequency.

3. Casualty Analysis

Visualizations showing reported fatalities and injuries by country, date, and available casualty category.

Purpose: Understand the distribution and development of reported human losses.

4. Geographical Analysis

An interactive map displaying recorded conflict events across affected geographical areas.

Purpose: Identify geographical patterns and locations with high concentrations of recorded events.

5. Event Type Analysis

A breakdown of recorded events according to their classifications.

Purpose: Understand the composition of conflict-related incidents.

6. Humanitarian Impact Analysis

Visualizations presenting available information about civilian impacts, displacement, and infrastructure damage.

Purpose: Explore the humanitarian consequences of the conflict.

7. Economic Impact Analysis

Charts comparing conflict developments with selected economic indicators, such as oil prices.

Purpose: Explore potential associations between conflict developments and economic market movements without assuming direct causation.

8. Interactive Filters

The dashboard will provide filters based on available data, including:

- Date Range.
- Country.
- Event Type.
- Casualty Category.
- Geographical Region.

3.6 Technical Requirements

The project will use the following technologies:

Technology	Purpose
Python	Data processing and analysis
Pandas	Data cleaning and manipulation
NumPy	Numerical calculations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Plotly	Interactive charts
Power BI	Interactive dashboard development
Jupyter Notebook	Exploratory data analysis and documentation
GitHub	Version control and project collaboration

3.7 Data Requirements

The project requires the following data categories:

Data Category	Required Information	Potential Source
Conflict Events	Event date, location, and event type	ACLED, IranWar.ai
Casualties	Reported fatalities and injuries	ACLED, UN reports, IranWar.ai
Geographical Data	Country, region, and coordinates	ACLED, IranWar.ai
Humanitarian Data	Displacement and infrastructure damage	UN/OCHA
Economic Data	Oil prices and selected market indicators	Public financial data providers

Each dataset will be reviewed for its date coverage, definitions, completeness, and reliability before integration.

3.8 Acceptance Criteria

The project will be considered successfully completed when:

1. Relevant datasets have been collected and documented.
2. Data cleaning and preprocessing have been completed.
3. Exploratory data analysis has been performed.
4. The main analytical indicators have been calculated.
5. Required visualizations have been developed.
6. The Power BI dashboard is interactive and functional.
7. Data sources and analytical limitations are clearly documented.
8. The final project files and documentation are uploaded to GitHub.

Conclusion

ConflictScope aims to demonstrate how data analysis and visualization can transform complex conflict-related information into clear, accessible, and meaningful insights. By combining selected public datasets, rigorous data cleaning, exploratory analysis, and an interactive Power BI dashboard, the project provides a structured view of military events, reported casualties, humanitarian impacts, and economic indicators, while clearly documenting data sources, definitions, and limitations.

GitHub Repository: https://github.com/darealbomo7/US-Iran-War-Data-Analysis